



PHASE 3 • POCKETSMART AI

Project Design Phase

Designing the system as connected layers: interface, backend, AI, and data sources.

Project Design Phase

PHASE PURPOSE

The design phase translates requirements into a modular technical structure. The supplied guide describes a web frontend, backend service layer, Gemini AI integration, and third-party platform data sources.

1 Frontend

HTML/CSS + JavaScript/Jinja2 interface for planner forms and recommendation display.

2 Backend

FastAPI implementation documented for routing, authentication, CORS, sessions and structured data.

3 Gemini AI

Gemini 1.5 Flash Pro interprets budgets, preferences, text and optional images.

4 Sources

Platform-oriented recommendations reference sources such as Amazon, Flipkart, IKEA, Zomato and related services.

PROCESS SNAPSHOT



COMPONENT ARCHITECTURE

Presentation Layer	Home page, planner forms, authentication pages, dashboard and recommendation cards.
Application Layer	Planner routes such as /generate-home, /generate-party and /generate-jewelry.
Intelligence Layer	Gemini prompts and recommendation logic for each domain.
Session Layer	Authentication, JWT token flow, session information and personalized data.

DESIGN PRINCIPLE

- Separate routes, services/recommendation logic and models for maintainability.
- Use structured inputs/outputs so the frontend and AI layer communicate predictably.
- Design the UI around three clearly separated planner journeys while retaining one PocketSmart identity.

PHASE TAKEAWAY

The design creates a modular architecture where UI, backend, AI services and data sources have clear responsibilities.

Source basis: uploaded PocketSmart AI reference guide. The guide documents the project's planners, architecture, workflow, milestones, development, testing and UI modules. The supplied YouTube link was not directly accessible in this environment, so no unsupported video-specific claims were added.